Body Control HLD

Synthetic AUTOSAR-inspired High-Level Design Knowledge Document

Important: This document is fictional, independently created for academic evaluation, and contains no confidential OEM data or reproduced AUTOSAR standard text.

Field	Value
Document ID	SYN-HLD-BC-001
Version	1.0
Status	Approved synthetic baseline
Classification	Synthetic academic test data
Intended use	RAG, OCR, extraction, comparison, and evaluation
Authority	Not valid for production design or vehicle release

1. Body control overview

The body control domain coordinates exterior lighting, door status, central locking, wiper control, and cabin illumination.

2. Component inventory

ID	Component	Role	Provides	Requires
SWC-BC-001	ExteriorLightingManager	Controls headlamp and indicator requests	If_LightingStatus	If_LightSwitch, If_VehicleSpeed
SWC-BC-002	DoorStateManager	Aggregates four door-ajar inputs	If_DoorState	If_DoorSensors
SWC-BC-003	CentralLockingManager	Commands lock actuators and reports status	If_LockStatus	If_DoorState, If_LockRequest
SWC-BC-004	WiperControlManager	Selects wipe state and interval	If_WiperStatus	If_WiperSwitch, If_RainLevel
SWC-BC-005	CabinLightingManager	Controls courtesy lighting	If_CabinLightStatus	If_DoorState, If_AmbientLight

3. Signals and validity

Signal	Source component	Interface	Range / states	Invalid behaviour
DoorAjar_FL	DoorStateManager	If_DoorState	OPEN/CLOSED/UNKNOWN	Publish UNKNOWN
AllDoorsClosed	DoorStateManager	If_DoorState	TRUE/FALSE	FALSE after timeout
LockState	CentralLockingManager	If_LockStatus	LOCKED/UNLOCKED/TRANSITION	TRANSITION for max 2 s
WiperState	WiperControlManager	If_WiperStatus	OFF/INT/LOW/HIGH	OFF
CabinLightLevel_pct	CabinLightingManager	If_CabinLightStatus	0..100	20% fallback

4. Functional relationships

DoorStateManager publishes If_DoorState to CentralLockingManager and CabinLightingManager. CentralLockingManager inhibits auto-lock when any door state is OPEN or UNKNOWN. CabinLightingManager commands 100% courtesy lighting when a door transitions from CLOSED to OPEN.

5. Controlled inconsistency

The component is named CabinLightingManager in the component inventory but is intentionally called CabinLightManager in the maintenance note below. The mismatch is included for terminology-consistency evaluation.

5.1 Maintenance note

CabinLightManager shall be inspected when CabinLightLevel_pct remains above 20% for more than 60 seconds after all doors are closed.

6. Review limitations

No physical pin mapping or network frame mapping is represented. Findings require engineer review.